



ZStar

Reproducible polarization, Born effective charge, phonon, infrared, Raman, and dielectric-response workflows.

PyPI v0.2.0

Python 3.9+

License GPL-3.0

English | 简体中文 | English PDF | 中文 PDF

Overview

ZStar is a Python workflow toolkit that connects ABACUS/PyATB, VASP, CP2K, Quantum ESPRESSO, and Phonopy calculations to physically auditable polarization and dielectric-response results. Its main task is to turn atomic response data into symmetry-consistent Born effective charge (BEC) tensors, then use those tensors for phonon, infrared (IR), dielectric, Raman, and molecular-dynamics (MD) analysis.

The toolkit keeps every stage visible: structures, solver inputs, band-gap gates, polarization values, charge-density data, tensor reconstruction reports, spectra, and progress records remain available for inspection and restart.

The numerical checks used for the current release are summarized in [docs/validation.md](#).

Main capabilities

- Forward and central finite-difference BEC calculations.
- Symmetry reduction, full-cell tensor reconstruction, and acoustic-sum-rule correction.
- A serial, resumable `0.no-move -> displaced structures` execution model.
- Reuse of the converged `0.no-move` charge density for every displacement.
- One-time insulating-state validation after the reference SCF.
- Shell, Slurm, and Torque driver generation.
- Automatic compatibility with legacy and direct-static-response PyATB versions.
- A serial, resumable CP2K backend for Berry-phase BEC tensors and native APT checks.
- Three-dimensional, hybrid two-dimensional, and hybrid one-dimensional polarization/BEC analysis.
- Phonon generation, post-processing, mode classification, IR spectra, Raman spectra, and dielectric response.

- MD dipole-fluctuation dielectric analysis using fixed or frame-dependent BEC tensors.
- Auxiliary electrostatic-potential analysis for slabs and polar materials.
- A packaged, standards-compliant Agent Skill with JSON workspace preflight.
- A versioned calculator-neutral response schema and backend plugin registry.
- Native Quantum ESPRESSO DFPT collection for molecular and bulk BEC/IR data.

The calculator-independent interface, physical `dim=0/1/2/3` contract, QE workflow, density adapters, Phonopy interchange, polarized Raman, optical constants, and external MD BEC providers are documented in the [calculator-independent guide](#).

Physical Scope

Three-dimensional crystals

For a periodic 3D crystal, ZStar evaluates

$$Z^*(\mathbf{kappa}, \alpha, \beta) = \Omega/e * dP_{\alpha} / du(\mathbf{kappa}, \beta)$$

from Berry-phase polarization differences. Polarization branches are matched modulo the polarization quantum before the finite difference is taken.

One-dimensional wires and nanowires

For a `z`-periodic wire, ZStar combines PYATB Berry polarization along `z` with real-space ABACUS cube dipoles along the nonperiodic `x/y` directions. It reports vacuum-independent BEC tensors and `Angstrom^2` line polarizabilities, and supports Gamma-point IR and Raman spectra. A bulk NAC is explicitly rejected because finite-wavevector polar phonons require a genuine 1D Coulomb cutoff. See the [one-dimensional workflow](#).

Two-dimensional slabs

A slab requires separate treatment of in-plane and out-of-plane response:

- **In-plane polarization columns:** Berry-phase polarization is used while the full supercell remains insulating.
- **Out-of-plane polarization column:** the total slab dipole is integrated from the ABACUS charge-density cube, including ionic and electronic contributions.
- **Normalization:** in-plane BECs are made independent of vacuum height through the usual volume factor; 2D dielectric spectra are reported as sheet polarizability unless an effective thickness is supplied.

Canonical ZStar tensors store atomic displacement/force as rows and polarization/electric field as columns. Accordingly, a complete 2D BEC calculation needs `x`, `y`, and `z` displacements. The default `zstar gen --dim 2` workflow generates all three. The current hybrid implementation requires the slab normal to align with Cartesian `z`; a tilted slab is rejected explicitly.

One reference/displaced cube pair can be audited independently:

```
zstar polar2d --reference-cube reference.cube \  
  --displaced-cube atom_zplus.cube \  
  --displacement 0.01 --outdir slab_dipole_check
```

This writes the planar charge redistribution, dipole finite difference, effective charge, diagnostics, and PNG/PDF/SVG plots.

Installation

ZStar requires Python 3.9 or newer.

Install the released package:

```
pip install -U zstar
```

Or install a local checkout:

```
git clone https://github.com/xdzhu/zstar.git  
cd zstar  
pip install .
```

External programs are required only for the corresponding workflows:

- ABACUS for SCF, charge density, force, and sparse-matrix calculations.
- PyATB for Berry-phase polarization, band checks, and electronic dielectric response.
- Phonopy for displacement generation and phonon post-processing.
- CP2K for the optional CP2K finite-displacement BEC backend.
- VASP for native bulk BEC and mode-displaced dielectric-response workflows.
- Quantum ESPRESSO for the optional native DFPT BEC, dielectric, and IR route.

Verify the command:

```
zstar --version  
zstar --help
```

Agent Skill

Install the bundled, standards-compliant `$run-zstar-workflows` skill after installing ZStar:

```
zstar agent-skill install  
zstar agent-skill preflight --root . --lane bec --dim bulk
```

Open a new agent session after installation. Use `--force` to refresh the skill after upgrading ZStar, or `--dest /path/to/skills` for a custom compatible skill directory. The skill encodes dimensional conventions, reference-first execution, restart behavior, scheduler authorization boundaries, and artifact-based completion checks. See [docs/agent_skill.md](#).

Born Charge Workflow

1. Generate the reference and displacement folders

Run this in a directory containing `STRU`:

```
zstar gen --stru STRU --pyatb --method forward --force
```

For a 2D slab:

```
zstar gen --stru STRU --dim 2 --pyatb --method forward --force
```

For a `z`-periodic wire:

```
zstar gen --stru STRU --dim 1 --pyatb --method central --force
```

The generated tree starts with `0.no-move`, followed by atom/direction folders such as `1.Ti/x+`. No per-displacement scheduler script is required.

Useful generation options:

Option	Meaning
<code>--method forward</code>	central
<code>--reduce</code> / <code>--all</code>	Symmetry-reduced atoms (default) or every atom.
<code>--move "x y z"</code>	Explicit displacement directions.
<code>--displacement 0.01</code>	Finite-displacement half-step in Angstrom.
<code>--dim 1</code>	2
<code>--input-mode abacus</code>	pyatb
<code>--input_sets FILES</code>	Extra files or directories copied into generated tasks.

2. Run the serial, resumable calculation

Local shell execution:

```
zstar workflow run --root . --dim 3 \  
  --abacus-command "mpirun -np 1 abacus" \  
  --pyatb-command "mpirun -np 1 pyatb" \  
  --omp-threads 28
```

For a wire or slab, use `--dim 1` or `--dim 2`, respectively. For low-dimensional polarization/BEC runs, ZStar writes `out_chg 1 10` so real-space transverse dipole differences are not limited by ABACUS cube rounding.

The default execution order is:

1. Run `0.no-move` SCF and save charge density and sparse matrices.
2. Generate a normal PyATB high-symmetry band path with `pyatb_input --band`.
3. Stop before any displacement if the reference is metallic.
4. Calculate reference polarization and electronic dielectric response.
5. Copy the reference charge cube/restart into each target `OUT.<suffix>/.`
6. Run every displacement serially and calculate its polarization.
7. Record stage state under `.zstar/` so an interrupted run can resume.

The regular band path is the default lightweight gate. It can detect a gap closure on the sampled path but cannot exclude an off-path metallic pocket. A denser MP-grid check is explicit:

```
zstar workflow run --gap-mode mp --mp-density 0.08
```

Inspect progress at any time:

```
zstar workflow status
```

3. Generate scheduler-specific drivers

Shell:

```
zstar workflow script --backend shell --dim 3
```

Slurm:

```
zstar workflow script --backend slurm --dim 3 \  
  --queue compute --cpus-per-task 28 --walltime 24:00:00
```

Torque/PBS:

```
zstar workflow script --backend torque --dim 3 \  
  --queue batch --cpus-per-task 28 --walltime 24:00:00
```

Backend-aware defaults use `mpirun -np N` for shell/Torque and `srun --ntasks=N` for Slurm. Add `--dry-run` to validate the generated script, environment, ordering, and resume records without launching a calculation. The three backends and their scheduler acceptance checks are recorded in [docs/validation.md](#).

Add `--submit` only when the generated script has been reviewed and the active environment provides the required executables.

4. Collect polarization and construct BEC tensors

Three-dimensional:

```
zstar deal --dim 3 --method forward --pyatb
```

Two-dimensional hybrid treatment:

```
zstar deal --dim 2 --method forward --pyatb
```

One-dimensional hybrid treatment for a **z**-periodic wire:

```
zstar deal --dim 1 --method central --pyatb
```

For central differences, use `--method central` consistently in both `gen` and `deal`.

Key outputs:

File	Meaning
<code>Z-BORN-reduced.out</code>	Raw tensors for explicitly calculated symmetry representatives.
<code>Z-BORN-symm.out</code>	Full-cell tensors reconstructed by symmetry and corrected by the acoustic sum rule.
<code>Z-BORN-reduced-neutral.out</code>	Reduced tensors after reconstruction and neutrality correction.
<code>BORN</code>	Electronic dielectric tensor plus Phonopy-order BEC tensors.
<code>BORN-for-phonopy.out</code>	Explicitly named copy of the Phonopy-compatible data.
<code>born_symmetry_report.json</code>	Machine-readable reconstruction and residual report.
<code>zstar_2d_bec.json</code>	Per-atom diagnostics for hybrid 2D BEC calculations.
<code>zstar_1d_bec.json</code>	Per-atom diagnostics for hybrid 1D BEC calculations.

CP2K BEC Backend

For a three-dimensional, insulating Gamma-point CP2K input, ZStar can build BEC tensors directly from periodic Berry-phase dipoles:

```

zstar cp2k-bec prepare --input input.inp --root cp2k_bec \
  --method central --displacement 0.005
zstar cp2k-bec run --root cp2k_bec --cp2k-command cp2k.smp \
  --omp-threads 20 --data-dir /path/to/cp2k/data
zstar cp2k-bec collect --root cp2k_bec

```

The reference wavefunction is reused by every displacement, and interrupted runs resume from `.zstar/cp2k_bec_state.json`. CP2K 2025.2+ native `APT_FD` results can be generated and compared through `cp2k-bec native` and `cp2k-bec compare`. See the [complete CP2K guide](#) for tensor conventions, input restrictions, convergence checks, and direct-node validation.

VASP BEC Backend

ZStar can also drive VASP's native BEC implementations. Use `dfpt` for `LEPSILON` linear response or `finite-field` for `LCALCEPS`/PEAD:

```

zstar vasp-bec prepare --input-dir vasp_input --root vasp_bec --method dfpt
zstar vasp-bec run --root vasp_bec --vasp-command "mpirun -np 20 vasp_std"
zstar vasp-bec collect --root vasp_bec

```

The reference SCF is checked for a finite band gap before the response stage, and completed stages are resumable. The collector writes normalized JSON, ZStar tensors, and a Phonopy-compatible `BORN` file. The [complete VASP guide](#) documents finite-field safeguards, cluster scripts, tensor conventions, and the VASP 6.3.2 SiC validation.

Phonons and Dielectric Response

1. Generate phonon calculations

In a phonon working directory containing `STRU`, `KPT`, and an `INPUT` with `cal_force 1`:

```

zstar ph --stru STRU --dim "2 2 2" --symprec 1e-3

```

Run every generated `disp-*` force calculation with the local execution system. `zstar ph` does not require or duplicate an `abacus_x.sh`; if one is present it is copied only as an optional convenience.

2. Post-process forces and classify Gamma modes

```

zstar postph
zstar irrep --file irreps.yaml --mode db

```

For non-analytical corrections, copy the BEC workflow's `BORN` into the phonon directory before post-processing:

```
cp ../polar/BORN .
zstar postph --nac
```

3. Static and frequency-dependent dielectric response

Copy the full tensors as well:

```
cp ../polar/BORN .
cp ../polar/Z-BORN-symm.out .
```

Static response:

```
zstar calc --qpoints qpoints.yaml --born Z-BORN-symm.out \
--dielectric BORN --dim 3
```

Frequency-dependent response:

```
zstar freq --qpoints qpoints.yaml --born Z-BORN-symm.out \
--dielectric BORN --dim 3 --plot
```

Modes below 5 cm⁻¹ are excluded by default; change this with `--acoustic-cutoff`.

For 2D, omit `--thickness` to obtain a vacuum-independent sheet polarizability in angstroms:

```
zstar calc --qpoints qpoints.yaml --born Z-BORN-symm.out \
--dielectric BORN --dim 2
```

Provide `--thickness` `ANGSTROM` only when an effective 3D dielectric tensor is desired.

Infrared Spectrum

Calculate mode effective charges, oscillator strengths, broadened IR intensity, and dielectric/sheet response:

```
zstar ir --qpoints qpoints.yaml \
--born Z-BORN-symm.out --dielectric BORN \
--dim 3 --broadening 10 --outdir ir_spectrum
```

Select modes explicitly when needed:

```
zstar ir --modes "4,5,8-10" --outdir ir_selected
```


Typical outputs are `ir_modes.csv`, `ir_spectrum.dat`, `ir_response_real.dat`, `ir_response_imag.dat`, `ir_spectrum.png`, `ir_spectrum.pdf`, `ir_spectrum.svg`, and `ir_summary.json`.

Raman Spectrum

ZStar obtains non-resonant Raman tensors by central finite differences of the electronic dielectric response along Gamma-point normal coordinates.

1. Prepare mode displacements

```
zstar raman prepare --stru STRU --qpoints qpoints.yaml \  
  --modes "4-12" --amplitude 0.02 --outdir raman \  
  --copy INPUT-scf --copy KPT
```

The amplitude is a normal-coordinate displacement in `angstrom * sqrt(amu)`.

2. Run, collect, and plot

```
zstar raman run --raman-dir raman \  
  --reference 0.no-move --qpoints qpoints.yaml \  
  --dim 3 \  
  --abacus-command "mpirun -np 1 abacus" \  
  --pyatb-command "mpirun -np 1 pyatb" \  
  --omp-threads 28
```

The reference insulating gate is reused once; it is not repeated for every mode displacement. Each `plus/minus` stage reuses the reference charge density and records resumable state.

Separate operations are also available:

```
zstar raman status --raman-dir raman  
zstar raman collect --raman-dir raman --qpoints qpoints.yaml --dim 3  
zstar raman spectrum --raman-dir raman --qpoints qpoints.yaml --dim 3
```

For 2D, `--dim 2` converts the vacuum-dependent dielectric derivative to a sheet-susceptibility derivative using the cell height stored in the phonon data.

Molecular IR and Raman

For the complete physical convention, workflow, outputs, and benchmark, see [Molecular IR and Raman spectroscopy](#).

An isolated molecule is represented in a sufficiently large periodic cell. Prepare its positive-frequency vibrational modes with the same central normal-coordinate displacements used for Raman calculations:

```
zstar raman prepare --stru STRU --qpoints qpoints.yaml \
  --acoustic-cutoff 100 --amplitude 0.02 --outdir raman \
  --copy INPUT-scf --copy KPT
```

The complete resumable calculation produces both spectra:

```
zstar raman run --raman-dir raman --reference 0.no-move \
  --qpoints qpoints.yaml --dim 0 \
  --abacus-command "mpirun -np 1 abacus" \
  --pyatb-command "mpirun -np 1 pyatb" \
  --spectrum-outdir raman_spectrum --ir-outdir ir_spectrum
```

Each displaced SCF is evaluated with two lightweight PYATB stages. The static dielectric response is converted to a molecular polarizability derivative, $d\alpha/dQ = V/(4\pi) * d(\epsilon_r)/dQ$. Branch-wrapped Berry polarization is converted to a molecular dipole derivative, $d\mu/dQ = V * dP/dQ$. The normal-coordinate step Q is in $\text{angstrom} * \text{sqrt}(\text{amu})$.

Existing polarization results can be post-processed independently:

```
zstar ir --dim 0 --qpoints qpoints.yaml \
  --displacements raman --outdir ir_spectrum
zstar raman spectrum --dim 0 --qpoints qpoints.yaml \
  --raman-dir raman --outdir raman_spectrum
```

Molecular spectra are normalized for mode assignment and workflow validation. They are not reported as gas-phase integrated cross sections. Use a converged vacuum size, basis, displacement amplitude, and electronic-response grid for quantitative intensity work.

VASP and CP2K calculators

The calculator-neutral spectroscopy layer also supports VASP and CP2K:

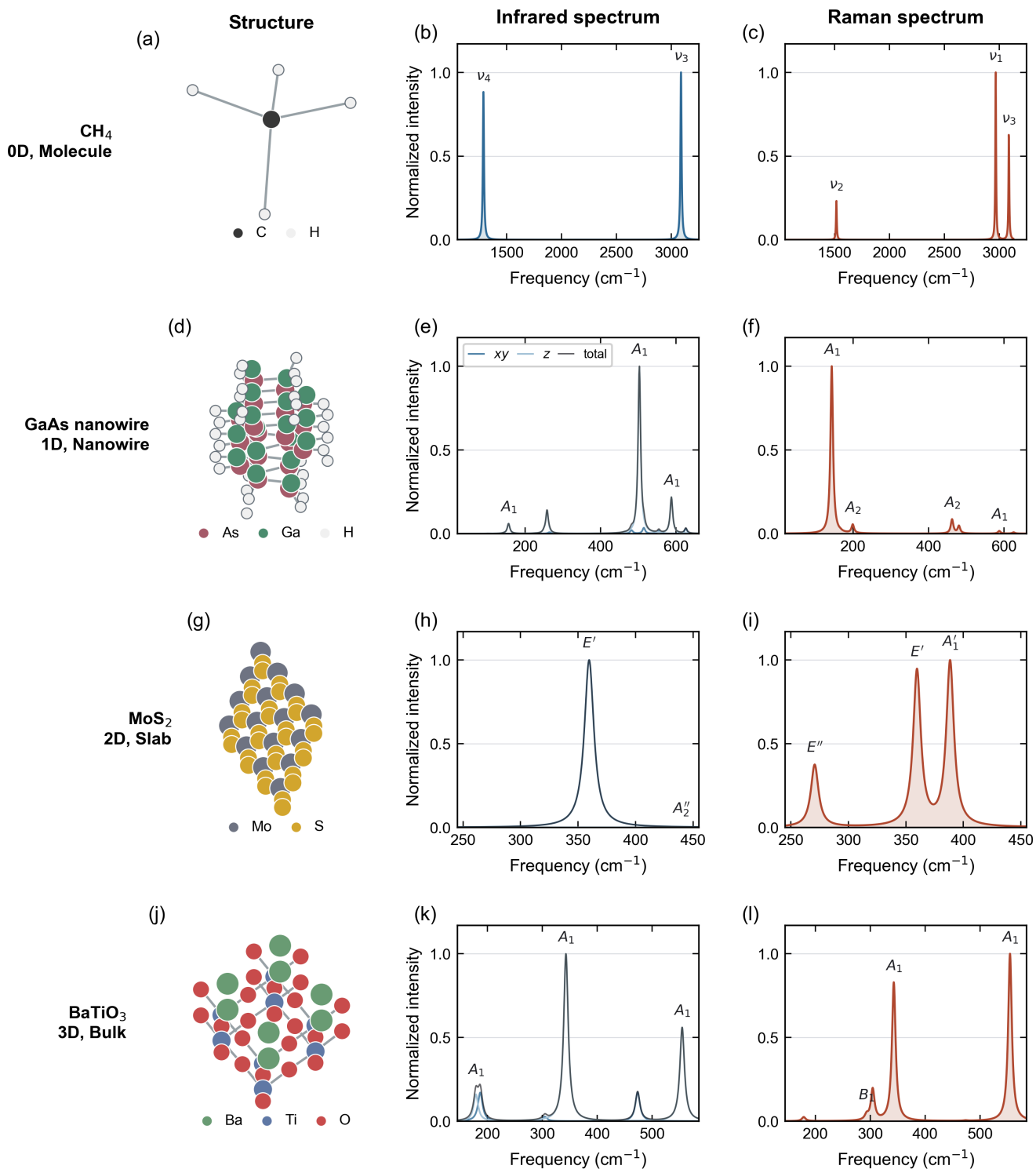
```
zstar spectra prepare --calculator vasp --input-dir vasp_input \
  --modes-xml phonon/vasprun.xml --root vasp_spectra --dim 3
zstar spectra run --root vasp_spectra --command "mpirun -np 20 vasp_std"
zstar spectra collect --root vasp_spectra

zstar spectra prepare --calculator cp2k --input h2o.inp \
  --root cp2k_spectra --dim 0
```

VASP uses central differences of native dielectric responses; CP2K uses native vibrational dipole and **LINRES/POLAR** intensities. See the [calculator spectroscopy guide](#).

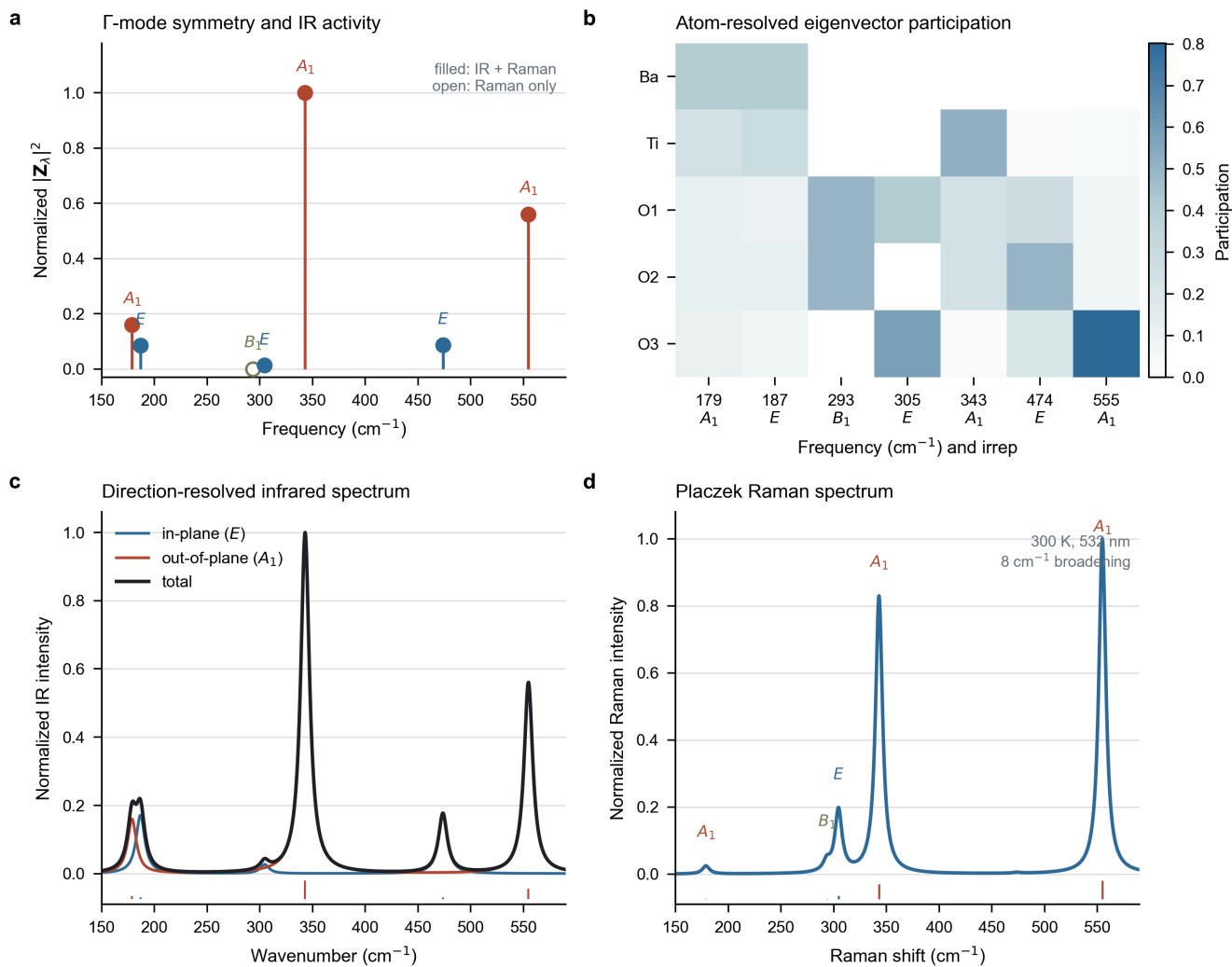
Representative Validation Figures

The compact source data, plotting script, vector files, and integrity manifest are archived in [docs/paper_figures](#).

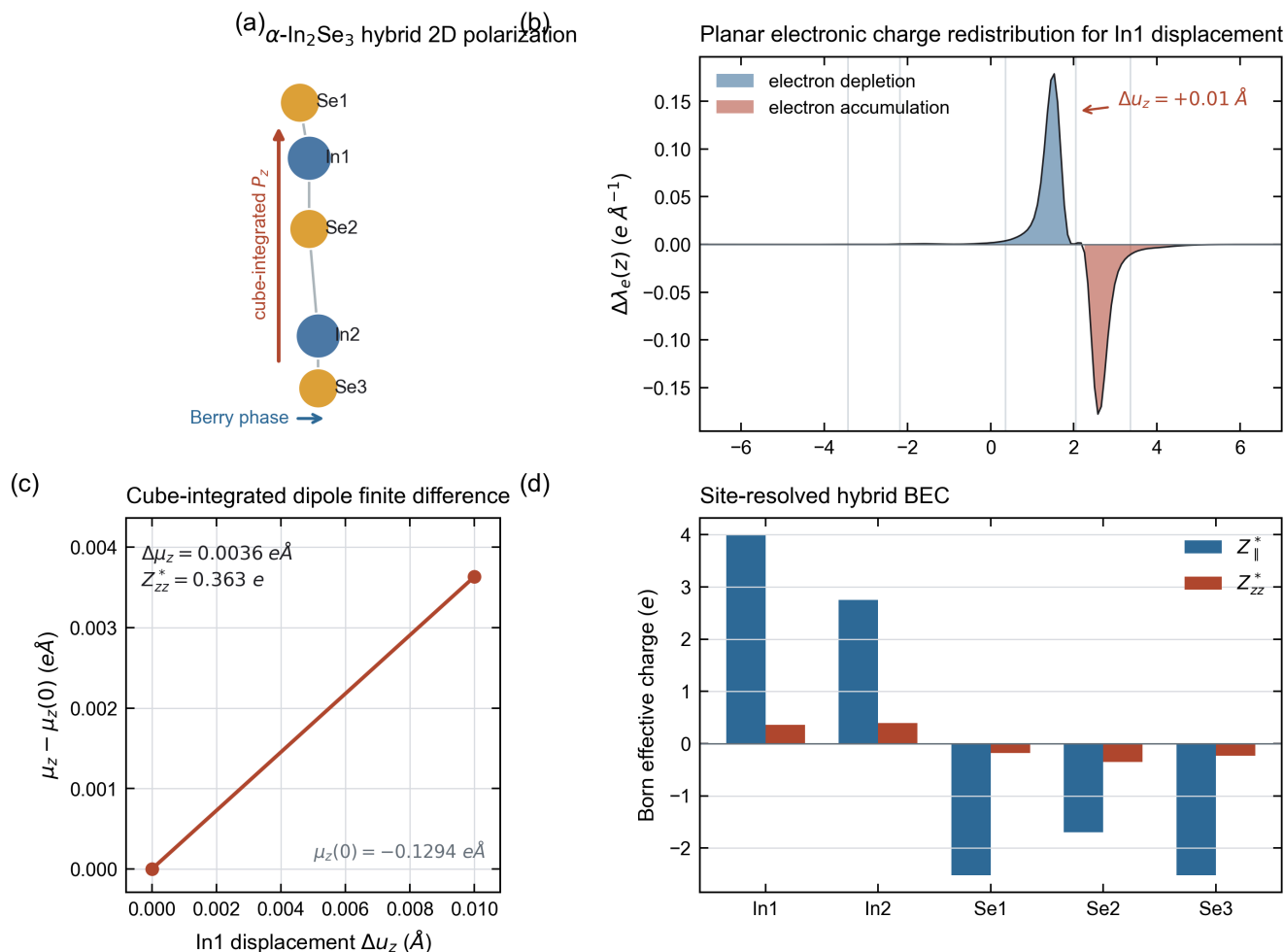


Each spectrum is normalized independently.

The four-row comparison contains completed **0D, Molecule**, **1D, Nanowire**, **2D, Slab**, and **3D, Bulk** calculations. The GaAs row retains all 68 positive-frequency IR modes and a disclosed ten-mode Raman subset covering all four **mm2** irreducible representations.



The BTO validation includes all ten positive-frequency optical modes and 20 completed Raman finite-difference response tasks. The **B1** mode at 293.38 cm^{-1} is Raman active and IR silent.



The In₂Se₃ validation displays the Berry-phase/cube-integral split and the actual planar charge redistribution for an out-of-plane In displacement.

MD + BEC Dielectric Response

zstar md does not prescribe how frame-dependent BECs are generated. They may come from:

- ZStar finite differences on selected snapshots.
- One fixed tensor set applied to every frame.
- An external charge/BEC model such as QNEP or another user workflow.

ZStar matches those tensors to the trajectory, reconstructs periodic displacements, builds the ionic dipole series, and evaluates its fluctuation susceptibility.

Fixed BEC tensors:

```
zstar md --dump dump.lammpstrj \
  --fixed-pec Z-BORN-symm.out \
  --electronic-dielectric BORN \
  --temperature 300 --type-map "1:Hf,2:Zr,3:O" \
  --outdir md_fixed
```

Frame-dependent tensors:

```
zstar md --dump dump.lampstrj \  
  --bec-dir bec_frames --bec-pattern "frame_{step}.npz" \  
  --electronic-dielectric BORN \  
  --temperature 300 --type-map "1:Hf,2:Zr,3:O" \  
  --start-step 200000 --stride-step 100 \  
  --outdir md_dynamic
```

The total static tensor is

```
epsilon_total = epsilon_infinity + chi_ionic
```

Outputs separate `chi_ionic.dat`, `epsilon_ionic.dat`, `epsilon_electronic.dat`, and `epsilon_total.dat`. If `--electronic-dielectric` is omitted, the identity tensor is used and the output is explicitly identified as `I + chi_ionic`.

PyATB Compatibility

ZStar probes the PyATB executable selected for the workflow:

- New builds with direct static response use `static_dielectric_only`.
- Older builds use a compact 0-30 eV optical grid at 0.1 eV spacing. This range was checked against the new direct-static intercept; the coarse spacing avoids an unnecessarily dense full optical spectrum.
- Both `static_dielectric_function.dat` and legacy `dielectric_function_real_part.dat` are accepted.

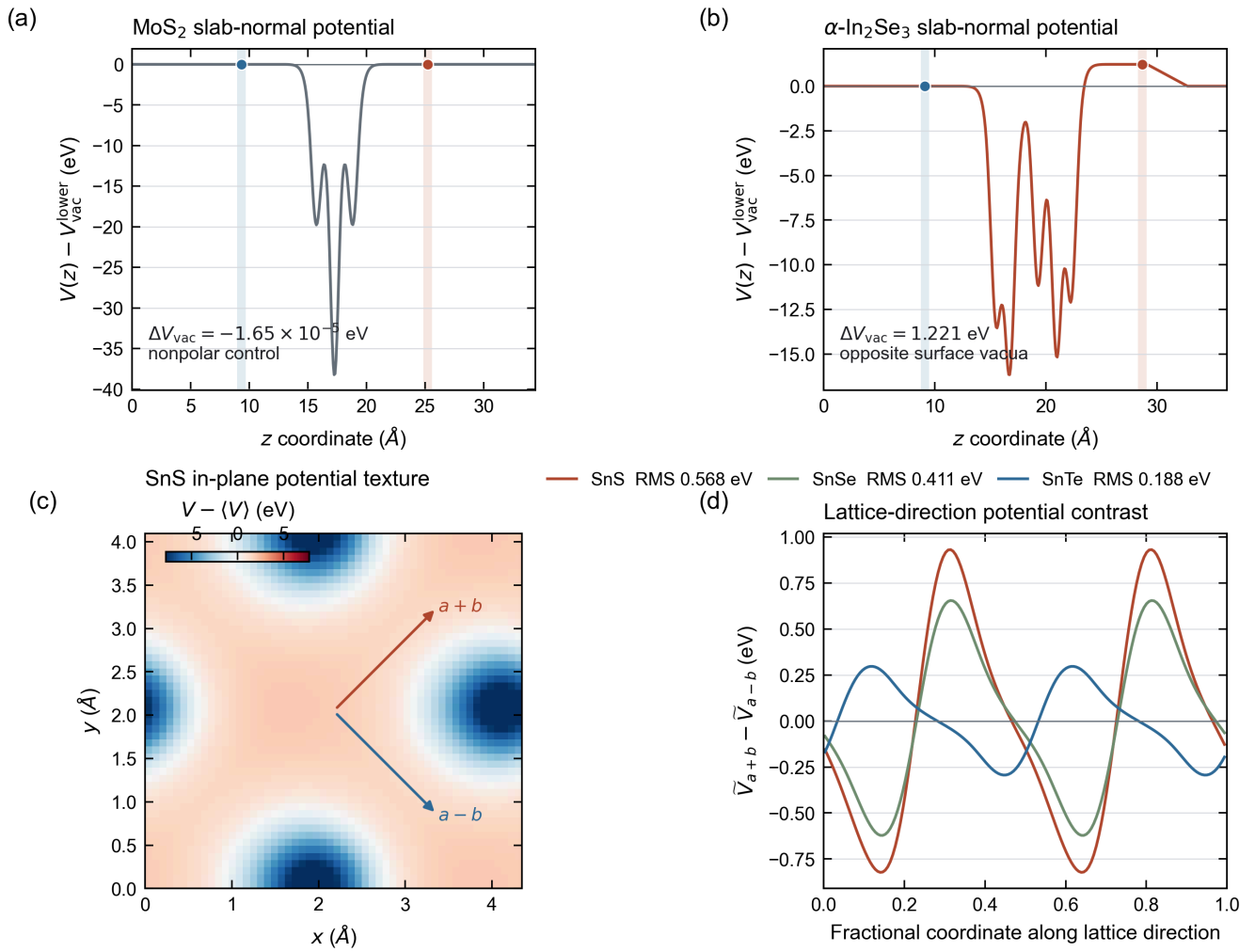
The selected mode and detected version are saved in `zstar_pyatb_compat.json`.

Electrostatic Potential

`zstar potential` (alias `zstar pot`) analyzes ABACUS `ElecStaticPot.cube` files:

```
zstar pot --cube OUT.ABACUS/ElecStaticPot.cube \  
  --axes z --plane xy --plane-average --tile 5 5 \  
  --vacuum-level --vacuum-sides --vacuum-window 0.75 \  
  --polar-arrow auto \  
  --outdir potential
```

It can generate axis profiles, tiled planar maps, directional averages, and one- or two-sided vacuum-level diagnostics. Local side windows avoid mixing a dipole-correction reset into a polar-slab surface plateau. In the representative tests, MoS₂ has `Delta V_vac = -1.65e-5 eV`, while alpha-In₂Se₃ has `Delta V_vac = 1.220812 eV`.



Commands, interpretation limits, and the SnS/SnSe/SnTe directional examples are collected in [docs/potential_examples.md](#).

Command Map

Command	Purpose
<code>zstar gen</code>	Generate reference and BEC displacement folders.
<code>zstar workflow run/status/script</code>	Run, inspect, or script the serial resumable workflow.
<code>zstar deal / born / polar</code>	Collect polarization and construct BEC tensors.
<code>zstar polar2d</code>	Audit a slab cube-pair dipole difference and out-of-plane BEC.
<code>zstar bornsym / symcheck</code>	Reconstruct or verify tensors by symmetry.
<code>zstar ph / postph</code>	Generate and post-process phonon tasks.
<code>zstar irrep</code>	Classify Gamma-point optical activity.
<code>zstar calc / freq</code>	Calculate static or frequency-dependent dielectric response.
<code>zstar ir</code>	Calculate mode charges and IR spectra.

Command	Purpose
<code>zstar raman</code>	Prepare, run, collect, and plot Raman finite differences.
<code>zstar spectra</code>	Run unified VASP or CP2K IR/Raman workflows.
<code>zstar md</code>	Combine an MD trajectory with fixed or frame-dependent BECs.
<code>zstar cp2k-bec</code>	Prepare, run, resume, collect, and validate CP2K BEC calculations.
<code>zstar db init/collect</code>	Create manifests and collect an auditable BEC/High-K database.
<code>zstar potential / pot</code>	Analyze electrostatic-potential cube files.
<code>zstar wyckoff / vasp</code>	Inspect Wyckoff positions or convert <code>STRU</code> .
<code>zstar agent-skill</code>	Install the Agent Skill or emit a JSON workspace preflight.

Repository and Release Policy

- `examples/` contains local validation data and is intentionally ignored.
- `dist/` and `build/` are local build products and are not committed.
- `job_scripts/` contains reusable scheduler templates and is version controlled.
- `docs/how_to_update_pypi.md` records the release procedure.

The relative logo works for authenticated viewers of a private GitHub repository. PyPI requires a public HTTPS image URL, so `README_PYPI.md` deliberately omits private images.

Citation and License

If ZStar supports published work, please cite the ZStar software article or repository release together with the underlying electronic-structure and lattice-dynamics programs used in the calculation.

Machine-readable citation metadata for this release are provided in `CITATION.cff`.

ZStar is distributed under the GNU General Public License v3.0.

Copyright (c) Xudong Zhu.